Yuanshuo Du

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Summary

Computer Vision and ML Engineering specialist with an MSc in Computer Science and deep learning coursework grounding. Hands-on expertise in CNN-based computer vision (ResNet50, R-CNN) and production ML system deployment. Daily practitioner of AI tools (Claude, RAG, n8n) for workflow automation, with a proven track record in CI/CD pipelines, containerized microservices, and Agile delivery.

Skills

ML / Deep Learning: Python, CNN, ResNet50, R-CNN, Deep Learning, TensorFlow, PyTorch, scikit-learn
Data / GIS: Python, R, PostgreSQL, MongoDB, GIS, Data Visualization, Spatial Analysis
AI Engineering: RAG, Claude API, FastAPI, LLM prompting, prompt engineering, n8n automation
MLOps / DevOps: Kubernetes, Docker, GitHub Actions CI/CD, Terraform, Linux, Go
Languages: Chinese (native), English (professional), Hungarian (conversational)

Technical Projects

Smart Parking Space Detection System — MSc CS Dissertation (Sep – Dec 2024)

CNN • ResNet50 • R-CNN • TensorFlow • Python • Kubernetes • Docker • Go • PostgreSQL

- Designed and trained CNN models (ResNet50, R-CNN) on satellite/aerial imagery to detect real-time parking space availability
- Deployed production microservices via Kubernetes/Docker with fully automated CI/CD – GitHub Actions cutting release cycles from 2 days to 15 minutes
- Built backend services in Go and Node.js; designed PostgreSQL schema for real-time parking availability data
- Co-authored and published research paper at *EduLearn25* Conference Proceedings

AI-Powered Q&A Platform — Mar 2024 – Sep 2024

RAG • Claude API • FastAPI • PostgreSQL • Docker • GitHub Actions

- Built Retrieval-Augmented Generation (RAG) pipeline using Claude API and FastAPI to answer user queries over large document corpora
- Implemented document chunking, embedding storage in PostgreSQL (pgvector), and semantic search retrieval pipeline
- Containerized the full stack with Docker; automated testing and deployment via GitHub Actions CI/CD
- Authored engineering documentation in GitBook for cross-team collaboration and onboarding

GIS for Public Amenities in Urban Areas — Jan 2025 – Jul 2025

GIS • Python • R • Data Visualization • Spatial Analysis

- Developed spatial analysis pipeline for identifying and visualizing public amenity distribution in urban areas using GIS
- Processed and analyzed large urban datasets; produced choropleth maps and accessibility metrics using Python and R
- Published as peer-reviewed paper in *EduLearn25* Conference Proceedings (Jul 2025)

Professional Experience

Software Engineer (Intern) — Stealth Startup (AI) — Mar 2024 – Oct 2024

- Containerized AI/ML inference services with Docker; documented deployment runbooks for production environments
- Instrumented model inference endpoints with Prometheus metrics; reduced monitoring setup time by 60% for new services

Information Technology Consultant — InsightSec, Shanghai — Mar 2021 – Dec 2022

- Designed and implemented CI/CD pipelines for enterprise clients across retail and pharmaceutical sectors – reduced manual deployment effort by 80%
- Applied Agile/Scrum across all engagements; managed backlogs in Jira, coordinated developers and QA teams

Education

MSc Computer Science — Technological University Dublin — Sep 2023 – Dec 2024

Deep Learning, Systems Architecture, Web App Architectures, Agile Methods

MSc Management (International Business) — University of Sheffield — Jun 2019 – Mar 2021

Agile Project Management, Accounting & Financial Management

Certifications & Publications

ISO 27001 Foundation – BSI (2021) ▪ CMI Level 7 Strategic Management & Leadership (2021) ▪ Google Analytics Advanced Certification ▪ *The Development of a GIS for Public Amenities in Urban Areas* – EduLearn25 Conference Proceedings (Jul 2025)